

Shri Aurobindo Institute of Technology, Indore

Department of CSE

**Basic Computer Engineering
(B.Tech.-205)**

Submitted By:-

Student Name:_____

Enrollment No:_____

Branch/Year/Semester:_____

Guided By:- Prof. Shweta Jaiswal

Signature

Software requirement

- 1. DOS**
- 2. MS office**
 - i) Ms word**
 - ii) Ms excel**
 - iii) Ms powerpoint**
 - iv) Ms access**
- 3. GCC**
- 4. Linux**

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S.No.	Objective	Date of Sub.	Remarks
1.	Study and practice of Internal & External DOS commands.		
2.	Study and practice of MS windows – Folder related operations, My-Computer, Window explorer, Control Panel,		
3.	Study and practice of Basic Linux Commands – ls, cp, mv, rm, chmod, kill, ps etc		
4.	Creation and editing of Text files using MS- word.		
5.	Creation and operating of spreadsheet using MS-Excel.		
6.	Creation and editing power-point slides using MS- power point		
7.	Creation and manipulation of database Table using SQL in MS-Access.		
8.	WAP to illustrate Arithmetic expressions		
9.	WAP to illustrate Arrays		
10.	WAP to illustrate functions		
11.	WAP to illustrate constructor & Destructor		
12.	WAP to illustrate Object and classes		
13.	WAP to illustrate Operator overloading		
14.	WAP to illustrate Function overloading		
15.	WAP to illustrate Derived classes & Inheritance		
16.	WAP to insert & delete element from the stack		
17.	WAP to insert & delete element from the queue		
18.	WAP to insert & delete element from the linked list		

PROGRAM NO: - 1

Topic: 1/DOS COMMANDS

PROBLEM DEFINITION:

Study and practice of Internal & External DOS commands.

OBJECTIVE:

To understand various basic commands of DOS.

ALGORITHM:

1. To Open MS DOS (For Windows Operating System)
2. Click on Start Button.
3. Click on programs.
4. Click on Accessories.
5. Click on command prompt.

Type `cd\` and press enter key on the command prompt.

After above step we will see `c :\>` on the screen. `C :\>` is called Dos Prompt/ System prompt/command prompt. It is pronounced as "c" prompt.

Type HELP for different commands then press Enter.

To close or quit from DOS (For Windows Operating System)

Type exit and press enter key.

For more information on a specific command, type HELP then command-name

For example: Help dir then press Enter

ASSOC	Displays or modifies file extension associations.
AT	Schedules commands and programs to run on a computer.
ATTRIB	Displays or changes file attributes.
BREAK	Sets or clears extended CTRL+C checking.
CACLS	Displays or modifies access control lists (ACLs) of files.
CALL	Calls one batch program from another.
CD	Displays the name of or changes the current directory.
CHCP	Displays or sets the active code page number.
CHDIR	Displays the name of or changes the current directory.
CHKDSK	Checks a disk and displays a status report.
CHKNTFS	Displays or modifies the checking of disk at boot time.
CLS	Clears the screen.

CMD	Starts a new instance of the Windows comand interpreter.
COLOR	Sets the default console foreground and background colors.
COMP	Compares the contents of two files or sets of files.
COMPACT	Displays or alters the compression of files on NTFS partitions.
CONVERT	Converts FAT volumes to NTFS. You cannot convert the current drive.
COPY	Displays a list of files and subdirectories in a directory. DISKCOMP Compares the contents of two floppy disks.
DISKCOPY	Copies the contents of one floppy disk to another.
DOSKEY	Edits command lines, recalls Windows commands, and creates macros.
ECHO	Displays messages, or turns command echoing on or off.
ENDLOCAL	Ends localization of environment changes in a batch file.
ERASE	Deletes one or more files.
EXIT	Quits the CMD.EXE program (command interpreter).
FC	Compares two files or sets of files, and displays the differences between them.
FIND	Searches for a text string in a file or files.
FINDSTR	Searches for strings in files.
FOR	Runs a specified command for each file in a set of files.
FORMAT	Formats a disk for use with Windows.
FTYPE	Displays or modifies file types used in file extension associations.
START	Starts a separate window to run a specified program or command.
SUBST	Associates a path with a drive letter.

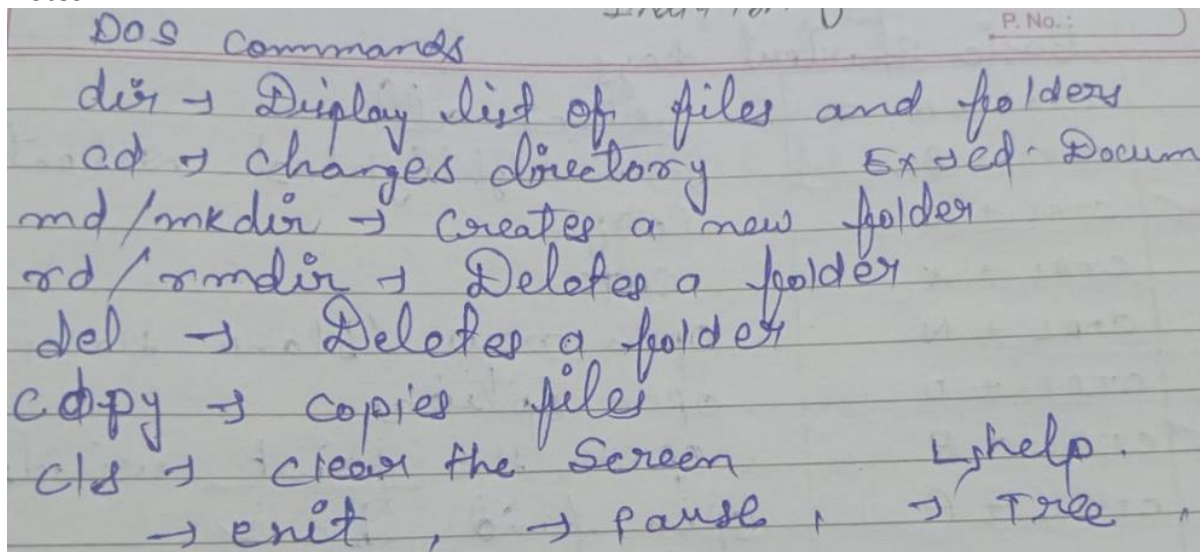
Input set

```
Microsoft Windows [Version 10.0.26200.8655]  
(c) Microsoft Corporation. All rights reserved.
```

Out put set:

```
For more information on a specific command, type HELP command-name  
ASSOC          Displays or modifies file extension associations.  
ATTRIB         Displays or changes file attributes.  
BREAK          Sets or clears extended CTRL+C checking.  
BCDEDIT        Sets properties in boot database to control boot loading.  
CACLS          Displays or modifies access control lists (ACLs) of files.  
CALL           Calls one batch program from another.  
CD             Displays the name of or changes the current directory.
```

Notes :



DOS Commands

- dir → Display list of files and folders
- cd → changes directory
- md/mkdir → creates a new folder
- rd/rmdir → Deletes a folder
- del → Deletes a folder
- copy → copies files
- cls → clear the Screen
- exit , → pause , → Tree

Ex: del. Docum
L:help.

DATE:

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PROGRAM NO: - 2

PROBLEM DEFINITION:

Topic: 2/MS WINDOWS

Study and Practice of MS windows – Folder related operations, My-Computer, window explorer, Control Panel.

OBJECTIVE:

Understanding MS windows – folder related operations, My computer, windows explorer, control panel.

ALGORITHM:

Windows

An operating system introduced by Microsoft Corporation in 1983. Windows is a multitasking graphical user interface environment that runs on MS-DOS-based computers (Windows 3x and Windows for Workgroups), and as a self-contained operating system for desktop computers (Windows 9x), workstations, (Windows NT Workstation, Windows 2000 Professional), and network servers (Windows NT Server, Windows NT Enterprise Edition, Windows 2000 Server, and Windows 2000 Advanced Server). Windows provides a standard graphical interface based on drop-down menus, windowed regions on the screen, and a pointing device such as a mouse.

Windows XP

Windows XP is an [operating system](#) produced by [Microsoft](#) for use on [personal computers](#), including home and business desktops, [laptops](#), and [media centers](#). It was first released in August 2001, and is the most popular version of [Windows](#), based on installed user base. The name "XP" is short for "eXPerience." Windows XP is the successor to both [Windows 2000](#) and [Windows Me](#), and is the first consumer-oriented operating system produced by Microsoft to be built on the [Windows NT kernel](#) and [architecture](#)

Files and Folders

A file is very much like a typed document that you might find on someone's desk or in a filing cabinet; it's an item that contains a collection of related information. On a computer, examples of files include text documents, spreadsheets, digital pictures, and even songs. Every picture you take with a digital camera, for example, is a separate file, and a music CD might contain a dozen individual song files.

Our computer represents files with icons. By looking at a file's icon, you can tell at a glance what kind of file it is.

A folder is little more than a container in which you can store files. If you put thousands of paper files on someone's desk, it would be virtually impossible to find any particular one

when you needed it. That's why people often store paper files in folders inside a filing cabinet. Arranging files into logical groups makes it easy to locate any particular file. Folders on our computer work exactly the same way.

Not only do folders hold files, but they also can hold other folders. A folder within a folder is usually called a subfolder. You can create any number of subfolders, and each can hold any number of files and additional subfolders.

To create a new folder

1. To open My Documents, click Start, and then click My Documents.
 2. Under File and Folder Tasks, click Make a new folder.
 3. A new folder is displayed with the default name, New Folder, selected.
 4. Type a name for the new folder, and then press ENTER.
- ☐ You can also create a new folder by right-clicking a blank area in a folder window or on the desktop, pointing to New, and then clicking Folder.

To move a file or folder

1. To open My Documents, click Start, and then click My Documents. If the file or folder, you want to move is not located in My Documents or its subfolders, use Search to find it.
2. To open Search, click Start, and then click Search.
3. Click the file or folder you want to move.
4. Under File and folder Tasks, click Move this file or Move this folder.
5. In Move Items, click the new location for the file or folder, and then click Move
6. To select a consecutive group of files, click the first file, press and hold down SHIFT, and then click the last file. To select files or folders in nonconsecutive order, press

Input:

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\anush_m>cd\

C:\>start
C:\>my documents
'my' is not recognized as an internal or external command,
operable program or batch file.
```

Output:

PROGRAM NO: -3

Topic: 3/LINUX

PROBLEM DEFINITION:

Study and practice of Basic Linux Commands – ls, cp, mv, rm, chmod, kill, ps etc.

OBJECTIVE:

To understand the various basic commands of Linux.

ALGORITHM:

1. Click on start then click on run. Type telnet "IP ADDRESS" .
 2. First enter username and password. For example username is "root".
 3. Black screen will appear.
 4. Type various commands listed below:
-

Command	Meaning	Description
Cat		Sends file contents to standard output. This is a way to list the Contents of short files to the screen. It works well with piping.
	Cat.bashrc	Sends the contents of the ".bashrc" file to the screen.
Cd		Change directory.
	Cd/home	Change the current working directory to /home.The "/" indicates relative to root, and no matter what directory you are in when you execute this command, the directory will be changed to "/home".
	Cd httpd	Change the current working directory to httpd, relative to the current location which is "/home/httpd".
	cd..	Move to the parent directory of the current directory. This command will make the current working "/home".
	Cd~	Move to the user's home directory which is "/home/username". The '~' indicates the users home directory.
Cp		Copy files
	Cp myfile yourfile	Copy the files "myfile" to the file "yourfile" in the current working directory. This command will create the file "yourfile" if it doesn't exist. It will normally overwrite it without warning if it exists.
	Cp -i myfile yourfile	with the "-i" option, if the file "yourfile" exists, you will

be prompted before it is overwritten.

if the directory "srcdir" to the directory "destdir" preserving links (-p option), file attributes (-p option), and copy recursively (-r option). With these options, a directory and all its contents can be copied to another directory.

Dd Disk duplicate. The man page says this command is to
Dd if=/dev/hdb1 "convert and copy a file", but although used by
more

ex

Of=/backup/ advanced users, it can be a very handy
command. The "if" means inputfile, "of" means output file.

O

DF Show the amount of disk space used on each mounted file
system.

d

Less Similar to more command, but the user can page up and

Less `textfile` and down through the file. The example displays the contents of `textfile`.

`Ln` creates a symbolic link to a file.

`Ln -s test symlink` creates a symbolic link named `symlink` that points to the file

Test typing "`ls -li test symlink`" will show the two files are different with different inodes. Typing "`ls -ltest are symlink`"

Will show that `symlink` points to the file `test`.

`Locate` A fast database driven file locate

`Slocate -u` This command builds the `slocate` database. It will take several minutes to complete this command. This command must be used before searching for files; however `cron` runs this command periodically on most systems.

`Locate whereis` List all files whose names contain the string "`whereis`".

Logout	Logs the current user off the system.
Ls	List files.
Ls	List files in the current working directory except those starting with. and only show the file name.
Ls -al	List all files in the current working directory in long listing format showing permissions, ownership, size, and time and date stamp.
More	Allows file contents or piped output to be sent to the screen one page at a time.

The “if” means inputfile, “of” means output file.

SDF the amount of disk space used on each mounted file system.

Less Similar to more command, but the user can page up and

Less textfile and down through the file. The example displays the contents of textfile.

Ln creates a symbolic link to a file.

Ln -s test symlink creates a symbolic link named symlink that points to the file

Test typing “ls -i test symlink” will show the two files are different with different inodes.

Typing “ls -ltest are symlink”

I Will show that symlink points to the file test.

Locate A fast database driven file locates.

This command builds the slocate database. It will take several minutes to complete this command. This command must be used before searching for files; however cron runs this command periodically on most systems.

Locate whereis List all files whose names contain the string “whereis”.

Logout Logs the current user off the system.

Ls List files.

Ls List files in the current working directory except those starting with. and only show the file name.

Ls -al List all files in the current working directory in long listing format showing permissions, ownership, size, and time and date stamp.

More Allows file contents or piped output to be sent to the screen one page at a time.

More
/etc/profile

List the contents of the “/etc/profile” file to the screen one page at a time.

Ls -al | more

performs a directory listing of all files and pipes the output of the listing through more. If the directory listing is longer than a page, it will be listed one page at a time.

Move or rename files.

Type exit to quit the linux

Input set:

```
user$ pwd
/tmp/user/home
user$ mkdir student
user$ cd student
user$ touch notes.txt
user$ ls
notes.txt
```

Output set:

Date

12/07/26.



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PROGRAM NO: - 4

Topic: 4/MS-WORD

PROBLEM DEFINITION:

Creation and editing of Text files using MS- word.

OBJECTIVE:

To understand the various features and formatting options in MS-word 2003.
Example: Build your resume in the MS-WORD.

ALGORITHM:

1. ClickonstartandthenclickonallprogramsandthenclickonMSOffice.
 2. OpenMS-WORD 2003& clickon filethen selectthe submenunew.
-

File: In this options available are save, open, close, print preview etc

- Edit: In this options available are paste, clear, select all, find, replace etc.
- View: In this options available are normal, print layout, toolbars, zoom etc.
- Insert: In this options available are picture, diagram, header footer etc.
- Format: In this options available are font, paragraph, numbering etc
- Tools: In this options available are word count, templates, spelling and grammar etc.
- Table: In this options available are draw table, insert, select, delete etc.
- Windows: In this options available.

Input set: **Resume**

Anushka Sharma

+91 9876543210

anushka@email.com

Gwalior, Madhya Pradesh

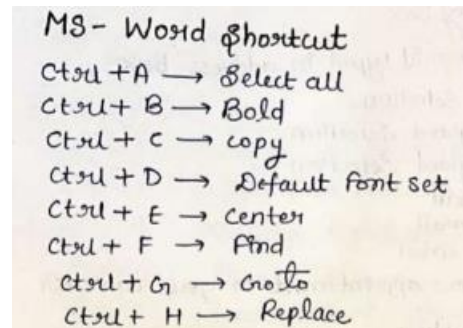
Educational Qualification

- B.Tech (1st Year) – RGPV University
- Higher Secondary (12th) – Passed
- High School (10th) – Passed

SKILLS: c, c++ basic, communication etc.

DATE 13/07/26.

Notes:



MS- Word shortcut
Ctrl + A → Select all
Ctrl + B → Bold
Ctrl + C → copy
Ctrl + D → Default font set
Ctrl + E → Center
Ctrl + F → Find
Ctrl + G → Go to
Ctrl + H → Replace

SALT Indore

PROGRAM NO:-5

Topic: 5/MS EXCEL

PROBLEM DEFINITION:

Creation and operating of spreadsheet using MS-Excel.

OBJECTIVE:

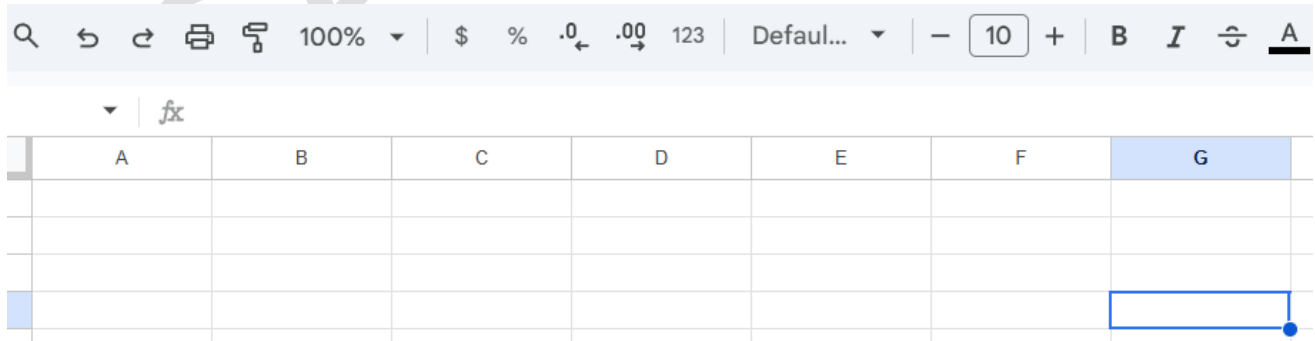
To understand the various features and formatting options in MS – Excel 2003.

Example: Create a spread sheet of all the students of a class and calculate total marks in five subjects, percentage and grade by applying suitable formula.

ALGORITHM:

1. Click on start and then click on all programs and then click on MS Office.
2. Open MS-EXCEL 2003 & click on the file then select the submenu new then click on the workbook.
3. We will see columns starting from A and rows starting from 1.
4. Give the heading Name to column A, Subject 1 marks to column B, Subject 2 marks to column C, Subject 3 marks to column D, Subject 4 marks to column E, Subject 5 marks to column F, Total marks to column G, Percentage to column H, Grade to column I.
5. Enter all the names of the students and their marks in the respective subjects.
6. In the total marks column we have to enter " = (B2+C2+D2+E2+F2)" in that cell and press enter key.
7. Now the total will be calculated in that cell, now point the mouse near the cell when "+" this appears left click and drag it down till the end of that column.
8. Doing this the formula will be applied to all the cells of that column.
9. For calculating percentage type this "= (G2/500)" in the percentage column and then repeat 7
10. For calculating grade use IF function .

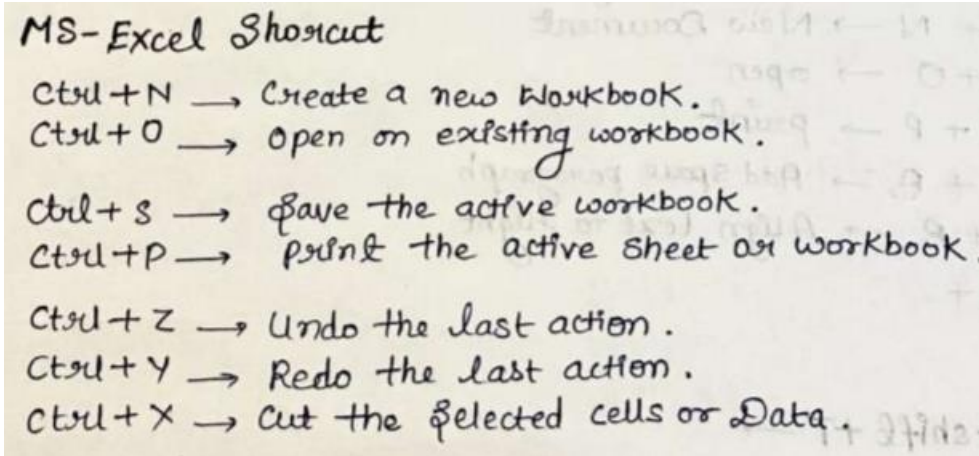
INPUT:



OUTPUT:

A	B	C	D	E	F	G	H	I
Name	sub 1	sub2	sub 3	sub 4	sub 5	total	percentage	Grade
Akash	56	78	90	56	67	347	69.4	C
Arjun	56	55	89	57	55	312	62.4	C
Aditi	78	99	87	57	33	354	70.8	B
Garima	0	88	76	88	54	306	61.2	C
Bhuvnesh	66	89	65	99	66	385	77	B

Notes:



PROGRAM NO: - 6

Topic: 6/MS POWER POINT

PROBLEM DEFINITION:

Creation and editing power-point slides using MS- power point.

OBJECTIVE:

To understand the features of MS power point.

Example: Create a presentation based on any topic.

ALGORITHM:

1. Click on start and then click on all programs and then click on MS Office 2003
 2. Open MS-POWER POINT 2003 & click on File then select New.
 3. Click on the New. A new slide will open.
 4. In menu bar you can see :
 - File: In this options available are save, open, close, print preview etc
 - Edit: In this options available are undo new slide, paste, clear, select all, find, replace etc.
 - View: In this options available are slide show, slide sorter, normal, toolbars, zoom etc.
 - Insert: In this options available are new slide, slide number, picture, diagram, header footer etc.
 - Format: In this options available are font, alignment, slide design, slide layout, etc
 - Tools: In this options available are auto correct options, spelling and compare and merge presentation etc.
 - Slide show: In this options available are view show, set up show, custom animations, animations schemes etc.
 - Windows: In this options available are new window, arrange all, cascade next pane etc.
 - Help: In this options available are power point help etc
-

INPUT SET:

Click to add title

Click to add subtitle

OUTPUT SET:

Boiler

Basic Mechanical Engineering
Submitted by: Anushka Sharma

1

What is a Boiler?

- A boiler is a closed vessel.
- It heats water to produce steam.
- Steam is used in industries and power plants.
- Fuel: coal, gas, oil or biomass.

2

Main Parts

- Boiler Shell
- Furnace
- Water Space
- Steam Space
- Fire Tubes
- Safety Valve
- Pressure Gauge
- Feed Pump

3

Working

- 1. Water enters the boiler.
- 2. Fuel burns in the furnace.
- 3. Heat converts water into steam.
- 4. Steam is supplied for work.

Types

- Fire Tube Boiler
 - Hot gases inside tubes.
 - Water outside tubes.
- Water Tube Boiler
 - Water inside tubes.
 - Hot gases outside tubes.

Applications

- Thermal Power Plants
- Textile Industry
- Sugar Industry
- Chemical Industry
- Food Processing
- Hospitals

PROGRAM NO: -7

Topic: 7/MS ACCESS

PROBLEM DEFINITION:

Creation and manipulation of database Table using SQL in MS-Access.

OBJECTIVE:

Understanding the SQL queries for creating and manipulating of database in MS Access.

Example: Create a employee database of a company.

ALGORITHM:

1. Go to Start > Programs > Microsoft Office > Microsoft Access 2003. You will see the gray screen with Access menu on top.
 2. Click File in the menu, choose New. The New File pane will open on the right-hand side of the screen.
 3. In the New File pane, select Blank Database option. This will open the File New Database dialog box.
 4. In the File New Database dialog box, type in a name for your database, navigate to the folder where you wish to save your file, and click Create.
-

CREATING A TABLE

1. After saving your database file you will be able to see the Database window. In the database window you will see three options for creating tables:
 - Create table in Design view.
 - Create table by using wizard.
 - Create table by entering data
2. If you choose the Create table in Design view option, you can name your fields, assign data type for each field, and format your fields.
3. Choosing the Create table by using wizard option allows you to create a table by following instructions in the Table Wizard dialog boxes.
4. If you decide to use the Create table by entering data option, you can enter data first, and then do all the formatting

FORMATTING A TABLE

Once you have created a table you can format it in Design View. Click on the Design View button in the toolbar.

SETTING PRIMARY KEY

1. The first step you should take while formatting your table is setting a Primary Key. The Primary Key, which is assigned to one of the fields, is a unique identifier of each record in a table. To set the Primary Key, do the following:
2. Select the field you want to be the Primary Key for your table. This is usually a number, for example employee id, a product serial number, etc.
3. Click the Primary Key button in the toolbar. This is now the unique identifier of the record.

INSERTING COLUMNS

1. If you need to insert a column into your table, i.e. if you need to add a field, you can do it from Design View.
2. Click the Design View button in the table grid you will see extra fields. Type in a title in a blank field.
3. Click on the Data Sheet View button. You will see a dialog box asking you to save the changes. Click OK.
4. Scroll all the way to the right to see the new column in your table.

FORMATTING DATA TYPE

In Design View you can also define the data type for each of the fields. For example, if you have a field that contains dollar values, you can define the data type for this field as Currency. The field will display the dollar sign and two decimal points.

To format the data type in Design View, follow these steps:

1. Click in the cell next to the field you wish to format the data type for in Data Type column.
2. Choose the desired type from the dropdown menu, Click on the Data Sheet View button. You will see a dialog box asking you to save the changes. Click OK.
3. Type in the data in the field; you will see how the formatting is applied.

INPUT SET:

ID	Book_name	writer	Price	Click to Add
(New)			₹ 0.00	

Output set:

ID	Book_name	writer	Price
1	Wings of Fire	APJ Abdul Kal	₹ 350.00
2	Rich Dad Poor Dad	Robert Kiyosa	₹ 499.00
3	Atomic Habits	James Clear	₹ 650.00
4	The Alchemist	Paulo Coelho	₹ 399.00

DATE: 12/07/26

Notes:

Access Specifier - A class can have member defined as public, private or protected to control accessibility.

PROGRAM NO: -8

Topic: 8/C++

PROBLEM DEFINITION:

WAP to illustrate Arithmetic expressions.

OBJECTIVE:

To understand the use of arithmetic expressions.

Example: WAP program to calculate grade of a student.

ALGORITHM:

1. Input 5 numbers of 5 subjects.
2. Add the numbers and calculate total number.
3. Divide total number by 5 to calculate %.
4. Check if % is in between 75 to 100, and then print honors.
5. Check if % is in between 60 to 75, and then print 1st division.
6. Check if % is in between 45 to 60, and then print 2nd division.
7. Check if % is less than 45, then print 3rd division.
8. Exit.

INPUT SET:

```
#include <iostream>
using namespace std;

int main()
{
    float s1, s2, s3, s4, s5;
    float total, percentage;

    // Input marks
    cout << "Enter marks of 5 subjects: ";
    cin >> s1 >> s2 >> s3 >> s4 >> s5;

    // Calculate total
    total = s1 + s2 + s3 + s4 + s5;

    // Calculate percentage
    percentage = total / 5;

    // Display result
    cout << "\nTotal Marks = " << total << endl;
    cout << "Percentage = " << percentage << "%" << endl;

    // Check division
    if (percentage >= 75 && percentage <= 100)
    {
        cout << "Result: Honors";
    }
    else if (percentage >= 60 && percentage < 75)
    {
        cout << "Result: 1st Division";
    }
    else if (percentage >= 45 && percentage < 60)
    {
        cout << "Result: 2nd Division";
    }
    else
    {
        cout << "Result: 3rd Division";
    }

    return 0;
}
```

OUTPUT SET:

```
Enter marks of 5 subjects: 73
71
65
60
55
```

```
Total Marks = 324
Percentage = 64.8%
Result: 1st Division
```

Date 12/07/26

PROGRAM NO: -9

Topic: 9 / C++

PROBLEM DEFINITION:

WAP to illustrate Arrays.

OBJECTIVE:

To understand the concept of array.

Example:WAPtoadd2linearlistsofnumbers.

ALGORITHM:

SALT,Indo

INPUT SET:

```
#include <iostream>
using namespace std;

int main()
{
    int arr[10];

    cout << "Enter 10 numbers:\n";
    for(int i = 0; i < 10; i++)
    {
        cin >> arr[i];
    }

    cout << "\nArray elements are:\n";
    for(int i = 0; i < 10; i++)
    {
        cout << arr[i] << " ";
    }

    return 0;
}
```

OUTPUT SET:

```
Enter 10 numbers:
2
4
8
9
2
|
0
3
7
1
6

Array elements are:
2 4 8 9 2 0 3 7 1 6
```

DATE:

12/07/26

PROGRAM NO: -
10

Topic: 10 / C++

PROBLEM DEFINITION:

WAP to illustrate functions.

OBJECTIVE:

To understand the features of functions.

Example: WAP to add and multiply with the help of function.

2.

ALGORITHM:

3. ~~Declare and define main function.~~

4. C++ 1. Declare a function with return type and parameters .It is of the following form:

return_type function_name (argument_list);

Input set:

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4
5 int add(int a, int b)
6 {
7     return a + b;
8 }
9
10 int multiply(int a, int b)
11 {
12     return a * b;
13 }
14
15 int main()
16 {
17     int x, y;
18
```

Output set:

```
Enter two numbers: 3
6
Addition = 9
Multiplication = 18
```

Date : 12/07/26

PROGRAM NO: -11A

Topic: 11A / C++

PROBLEM DEFINITION:

WAP to illustrate constructor & Destructor.

OBJECTIVE:

To explore the utilities of constructor.

WAP that add two numbers and display the result.

ALGORITHM:

1. Create a class.
 2. Create a constructor with the same name as a class name.
 3. Define that constructor
 - either inside the class
 - Or outside the class with scope resolution operator.
 4. Create an object of that class and it is initialized and constructor is invoked automatically.
-

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4
5 class Add
6 {
7     int a, b;
8     7
9
10 public:
11     Add()
12     {
13         cout << "Enter two numbers: ";
14         cin >> a >> b;
15
16         cout << "Sum = " << a + b << endl;
17     }
18 };
```

Output set:

```
Enter two numbers:  
5  
7  
Sum = 12
```

Date 12/07/26.

PROGRAM NO: -11b

Topic: 11B / C++

PROBLEM DEFINITION:

WAP to illustrate constructor & Destructor.

OBJECTIVE:

To explore the utilities for Destructor.

ALGORITHM:

4. Create a destructor with the same name as class name preceded by tilde.
 1. Create a class.
 2. Create a constructor with the same name as a class name.
 3. Define that constructor
-

Input se • either inside the class

```
#include <iostream>
using namespace std;

class Demo
{
public:
    // Constructor
    Demo()
    {
        cout << "Constructor is called." << endl;
    }

    // Destructor
    ~Demo()
    {
        cout << "Destructor is called." << endl;
    }
};

int main()
{
    Demo obj;    // Object created

    cout << "Inside main function." << endl;

    return 0;
}
```

Output set:

```
Constructor is called.  
Inside main function.  
Destructor is called.
```

DATE:12/07/26

PROGRAM NO: -12

Topic: 12/ C++

PROBLEM DEFINITION:

WAP to illustrate Object and classes.

OBJECTIVE:

To understand the concept of class and object.

ALGORITHM:

1. Create a class by the use of keyword 'class' followed by the class name.
2. Inside the class declare its public and private data members and member functions.

Input set:

```
#include <iostream>
using namespace std;

class Student
{
public:
    int roll;
    string name;

    void input()
    {
        cout << "Enter Roll No: ";
        cin >> roll;

        cout << "Enter Name: ";
        cin >> name;
    }

    void display()
    {
        cout << "\nRoll No: " << roll << endl;
        cout << "Name: " << name << endl;
    }
};

int main()
{
    Student s;    // Object of class

    s.input();
    s.display();

    return 0;
}
```

```
Enter Roll No: 251028
Enter Name: ANUSHKA SHARMA
```

```
Roll No: 251028
Name: ANUSHKA
```

Output set:

Date 12/07/26

PROGRAM NO: -13

Topic: 13/ C++

PROBLEM DEFINITION:

WAP to illustrate Operator overloading.

OBJECTIVE:

To understand the concept of overloading of an operators.

ALGORITHM:

1. Create a class that defines the data type that is to be used in the overloading operation.
2. Declare the operator function operator op () in the public part of the class. It may be either a public member of a class or a friend 3.function. Define the operator function to implement the required operation.

Input set :

```
#include <iostream>
using namespace std;

class Number
{
    int num;
public:
    void getData()
    {
        cout << "Enter number: ";
        cin >> num;
    }

    Number operator + (Number obj)
    {
        Number temp;
        temp.num = num + obj.num;
        return temp;
    }

    void display()
    {
        cout << "Sum = " << num << endl;
    }
};

int main()
{
    Number n1, n2, n3;

    n1.getData();
    n2.getData();

    n3 = n1 + n2;

    n3.display();

    return 0;
}
```

Output set

```
Enter number: 5
Enter number: 6
Sum = 11
```

Date 12/07/26

SALT, Indore

PROGRAM NO: -14

Topic: 14/ C++

PROBLEM DEFINITION:

WAP to illustrate Function overloading

OBJECTIVE:

To explore the utilities of function overloading.

WAP to calculate the sum of two numbers.

ALGORITHM:

1. Declare and define family of functions with one function name but different argument lists.

The correct function to be invoked is determined by checking the number and type of the argument but not on the function function call must matches the prototype having the same number and type of argumentsThen calls the appropriate function for execution.

2. Function overloading is the process of using the same name for two or more functions.
3. The correct function to be invoked is determined by checking the number and type of the argument but not on the function function call must matches the prototype having the same number and type of argumentsThen calls the appropriate function for execution.
4. Function overloading is the process of using the same name for two or more functions.
5. The correct function to be invoked is determined by checking the number and type of the argument but not on the function function call must matches the prototype having the same number and type of argumentsThen calls the appropriate function for execution.
6. Function overloading is the process of using the same name for two or more function.

```

#include <iostream>
using namespace std;

class Demo
{
public:
    void add(int a, int b)
    {
        cout << "Sum of Integers = " << a + b << endl;
    }

    void add(float a, float b)
    {
        cout << "Sum of Floats = " << a + b << endl;
    }

    void add(int a, int b, int c)
    {
        cout << "Sum of Three Integers = " << a + b + c << endl;
    }
};

int main()
{
    Demo d;

    d.add(10, 20);
    d.add(5.5f, 4.5f);
    d.add(10, 20, 30);

    return 0;
}

```

Input

```

Sum of Integers = 30
Sum of Floats = 10
Sum of Three Integers = 60

```

Output

Date 12/07/26

PROGRAM NO: -15

Topic: 15/ C++

PROBLEM DEFINITION:

WAP to illustrate Derived classes & Inheritance.

OBJECTIVE:

To explore the utilities of inheritance.

ALGORITHM:

1. Create a class as a base class.
2. declare data members and member functions access specifiers (public, private, protected)
3. member functions of the classCreate another class as derived class which is inheriting the features of base class.Create the objects of the derived classes using and base classes.
4. Call their member functions respectively.

Input set:

```
#include <iostream>
using namespace std;

// Base Class
class Student
{
public:
    int roll;
    void getData()
    {
        cout << "Enter Roll Number: ";
        cin >> roll;
    }
};

// Derived Class
class Result : public Student
{
public:
    int marks;

    void getMarks()
    {
        cout << "Enter Marks: ";
        cin >> marks;
    }

    void display()
    {
        cout << "\nRoll Number: " << roll << endl;
        cout << "Marks: " << marks << endl;
    }
};

int main()
{
    Result r;

    r.getData();
    r.getMarks();
    r.display();

    return 0;
}
```

Output set:

```
Enter Roll Number: 25
Enter Marks: 80

Roll Number: 25
Marks: 80
```


Notes

```
#include <iostream>
using namespace std;

class vehicle {
public:
    ()
    void start {
        cout << "Vehicle started" << endl;
    }
};

class Car : public vehicle {
public:
    void horn() {
        cout << "Car horn" << endl;
    }
};
```

PROGRAM NO: -16

Topic: 16/ DS

PROBLEM DEFINITION:

WAP to insert & delete element from the stack

OBJECTIVE:

To understand concept of stack.

ALGORITHM:

Adding into stack

```
procedure add(item : items);  
{add item to the global stack stack;  
top is the current top of stack  
and n is its maximum size}  
begin  
    if top = n then stackfull;  
    top := top+1;  
    stack(top) := item;  
end: {of add}
```

Deletion in stack

```
procedure delete(var item : items);  
{remove top element from the stack stack and put it in the item}  
begin  
    if top = 0 then stackempty;  
    item := stack(top);  
    top := top-1;  
end; {of delete}
```

Input set:

```
#include <iostream>
using namespace std;
#define MAX 5

class Stack
{
    int stack[MAX], top;
public:
    Stack()
    {
        top = -1;
    }

    void push(int item)
    {
        if (top == MAX - 1)
        {
            cout << "Stack Overflow\n";
        }
        else
        {
            top++;
            stack[top] = item;
            cout << item << " inserted into stack.\n";
        }
    }

    void pop()
    {
        if (top == -1)
        {
            cout << "Stack Underflow\n";
        }
        else
        {
            cout << stack[top] << " deleted from stack.\n";
            top--;
        }
    }

    void display()
    {
        if (top == -1)
        {
            cout << "Stack is empty\n";
        }
        else
        {
            cout << "Stack Elements: ";
            for (int i = top; i >= 0; i--)
            {
                cout << stack[i] << " ";
            }
            cout << endl;
        }
    }
};

int main()
{
    Stack s;
    s.push(10);
    s.push(20);
    s.push(30);
    s.display();
    s.pop();
    s.display();
    return 0;
}
```

Output set:

```
10 inserted into stack.
20 inserted into stack.
30 inserted into stack.
Stack Elements: 30 20 10
30 deleted from stack.
Stack Elements: 20 10
```

PROGRAM NO: -17

Topic: 17/ DS

PROBLEM DEFINITION:

WAP to insert & delete element from the queue

OBJECTIVE:

To understand concept of queue.

ALGORITHM:

Addition into a queue

```
procedure addq (item : items);  
{add item to the queue q}  
begin  
  if rear=n then queuefull  
  else begin  
    rear :=rear+1;  
    q[rear]:=item;  
  end;  
end; {of addq}
```

Deletion in a queue

```
procedure deleteq (var item : items);  
{delete from the front of q and put into item}  
begin  
  if front = rear then queueempty  
  else begin  
    front := front+1  
    item := q[front];  
  end;  
end; {of deleteq}
```

```

#define MAX 5
#include <iostream>
using namespace std;

#define MAX 5

class Queue
{
    int q[MAX];
    int front, rear;
public:
    Queue()
    {
        front = rear = -1;
    }

    void enqueue(int item)
    {
        if (rear == MAX - 1)
        {
            cout << "Queue Overflow\n";
        }
        else
        {
            if (front == -1)
                front = 0;
            rear++;
            q[rear] = item;
            cout << item << " inserted into queue.\n";
        }
    }

    void dequeue()
    {
        if (front == -1 || front > rear)
        {
            cout << "Queue Underflow\n";
        }
        else
        {
            cout << q[front] << " deleted from queue.\n";
            front++;
        }
    }

    void display()
    {
        if (front == -1 || front > rear)
        {
            cout << "Queue is Empty\n";
        }
        else
        {
            cout << "Queue Elements: ";
            for (int i = front; i <= rear; i++)
                cout << q[i] << " ";
            cout << endl;
        }
    }
};

int main()
{
    Queue q;

    q.enqueue(10);
    q.enqueue(20);
    q.enqueue(30);

    q.display();

    q.dequeue();

    q.display();

    return 0;
}

```

INPUT SET:

Output

```

10 inserted into queue.
20 inserted into queue.
30 inserted into queue.
Queue Elements: 10 20 30
10 deleted from queue.
Queue Elements: 20 30

```

Output set :

PROGRAM NO: -18

Topic: 18/ DS

PROBLEM DEFINITION:

WAP to insert & delete element from the linked list

OBJECTIVE:

To understand concept of linked list.

ALGORITHM:

Delete First Node

Delete First ():

Description: Here START is a pointer variable which contains the address of first node. ITEM is the value to be deleted.

1. If (START == NULL) Then [Check whether list is empty]
2. Print: Linked-List is empty.
3. Else
4. PTR = START
5. ITEM = START->INFO [Assign INFO of first node to ITEM]
6. START = START->LINK [START now points to 2nd node]
7. Delete PTR [Delete first node]
8. Print: ITEM deleted
[End of If]
9. Exit

Delete Last Node

Delete Last ():

Description: Here START is a pointer variable which contains the address of first node. PTR is a pointer variable which contains address of node to be deleted. PREV is a pointer variable which points to previous node. ITEM is the value to be deleted.

1. If (START == NULL) Then [Check whether list is empty]
2. Print: Linked-List is empty.
3. Else
4. PTR = START, PREV = START
5. Repeat While (PTR->LINK != NULL)
6. PREV = PTR [Assign PTR to PREV]
7. PTR = PTR->LINK [Move PTR to next node]
[End of While Loop]
8. ITEM = PTR->INFO [Assign INFO of last node to ITEM]
9. If (START->LINK == NULL) Then [If only one node is left]
10. START = NULL [Assign NULL]

1. ter vaPREV->LINK = NULL [Assign NULL to link field of second last node] [End of Step 9 If]
2. Delete PTR
3. Print: ITEM deleted
[End of Step 1 If]
4. Exit
- 5.

6. Delete Specific Node Delete

Specific ():

7. **Description:** Here START is a pointer variable which contains the address of first node. PTR is a pointer variable which contains address of node to be deleted. PREV is a pointer variable which points to previous node. ITEM is the value to be deleted.

1. If (START == NULL) Then [Check whether list is empty]
2. Print: Linked-List is empty.
3. Else If (START->INFO == ITEM) Then [Check if ITEM is in 1st node]
4. PTR = START
5. START = START->LINK [START now points to 2nd node]
6. Delete PTR
7. Else
8. PTR = START, PREV = START
9. Repeat While (PTR != NULL)
10. If (PTR->INFO == ITEM) Then [If ITEM matches with PTR->INFO]
11. PREV->LINK = PTR->LINK [Assign LINK field of PTR to PREV]
12. Delete PTR
13. Else
14. PREV = PTR [Assign PTR to PREV]
15. PTR = PTR->LINK [Move PTR to next node]
[End of Step 10 If]
[End of While Loop]
16. Print: ITEM deleted
[End of Step 1 If]
17. Exit

Insert First Node

Insert First ():

Description: Here START is a pointer variable which contains the address of first node. ITEM is the value to be inserted.

1. If (START == NULL) Then
2. START = New Node [Create a new node]
3. START->INFO = ITEM [Assign ITEM to INFO field]
4. START->LINK = NULL [Assign NULL to LINK field]
5. Else
6. Set PTR = START [Initialize PTR with START]
7. START = New Node [Create a new node]
8. START->INFO = ITEM [Assign ITEM to INFO field]

9. START->LINK = PTR [Assign PTR to LINK field]

[of If]

10. Exit

Insert Item in a Sorted List

Insert Sorted ():

Description: Here START is a pointer variable which contains the address of first node. PREV is a pointer variable which contains address of previous node. ITEM is the value to be inserted.

1. If (START == NULL) Then [Check whether list is empty]
2. START = New Node [Create a new node]
3. START->INFO = ITEM [Assign ITEM to INFO field]
4. START->LINK = NULL [Assign NULL to LINK field]
5. Else
6. If (ITEM < START->INFO) Then [Check whether ITEM is less then value in first node]
7. PTR = START
8. START = New Node
9. START->INFO = ITEM
10. START->LINK = PTR
11. Else
12. Set PTR = START, PREV = START
13. Repeat While (PTR != NULL)
14. If (ITEM < PTR->INFO) Then
15. PREV->LINK = New Node
16. PREV = PREV->LINK
17. PREV->INFO = ITEM
18. PREV->LINK = PTR
19. Return
20. Else If (PTR->LINK == NULL) Then [Check whether PTR reaches last node]
21. PTR->LINK = New Node
22. PTR = PTR->LINK
23. PTR->INFO = ITEM
24. PTR->LINK = NULL
25. Return
26. Else
27. PREV = PTR
28. PTR = PTR->LINK
- [End of Step 14 If]
- [End of While Loop]
- [End of Step 6 If]
- [End of Step 1 If]
29. Exit

Insert Last Node

Insert Last ():

Description: Here START is a pointer variable which contains the address of first node. ITEM is the value to be inserted.

1. If (START == NULL) Then [Check whether list is empty]

2. START == New Node
3. START->INFO = ITEM
4. START->LINK = NULL
5. Else
6. Set PTR = START [Initialize PTR with START]
7. Repeat While (PTR->LINK != NULL)
8. PTR = PTR->LINK [Until PTR reaches last node]
[End of While Loop]
9. PTR->LINK = New Node [Assign address of new node to PTR]
10. PTR = PTR->LINK [Move PTR to next node]
11. PTR->INFO = ITEM
12. PTR->LINK = NULL
[End of If]
13. Exit

Insert Specific Node

Insert Specific ():

Description: Here START is a pointer variable which contains the address of first node. NEW is a pointer variable which will contain address of new node. N is the value after which new node is to be inserted and ITEM is the value to be inserted.

1. If (START == NULL) Then
2. Print: Linked-List is empty. It must have at least one node
3. Else
4. Set PTR = START, NEW = START
5. Repeat While (PTR != NULL)
6. If (PTR->INFO == N) Then
7. NEW = New Node
8. NEW->INFO = ITEM
9. NEW->LINK = PTR->LINK
10. PTR->LINK = NEW
11. Print: ITEM inserted
12. ELSE
13. PTR = PTR->LINK
[End of Step 6 If]
[End of While Loop]
[End of Step 1 If]
14. Exit


```
// Delete Last Node
void deletelast()
{
    if (start == NULL)
    {
        cout << "Linked List is Empty\n";
    }
    else
    {
        start->next = NULL;
        cout << start->data << " deleted from last.\n";
        start = NULL;
    }
    PTR = start;
    while (PTR->LINK != NULL)
    {
        PTR = PTR->LINK;
    }
    PTR->LINK = New Node;
    PTR = PTR->LINK;
    PTR->INFO = ITEM;
    PTR->LINK = NULL;
}
void display()
{
    if (start == NULL)
    {
        cout << "Linked List is Empty\n";
        return;
    }
    Node *temp = start;
    while (temp != NULL)
    {
        cout << temp->data << " ";
        temp = temp->next;
    }
    cout << endl;
}
```

Insert Specific Node

Insert Specific (-):

Description: Here START is a pointer variable which contains the address of first node. NEW is a pointer variable which will contain address of new node. N is the value after which new node is to be inserted and ITEM is the value to be inserted.

1. If (START == NULL) Then
2. Print: Linked-List is empty. It must have at least one node
3. Else
4. Set PTR = START, NEW = START
5. Repeat While (PTR != NULL)
6. If (PTR->INFO == N) Then
7. NEW = New Node
8. NEW->INFO = ITEM
9. NEW->LINK = PTR->LINK
10. PTR->LINK = NEW
11. Print: ITEM inserted
12. ELSE
13. PTR = PTR->LINK
- [End of Step 6 If]
- [End of While Loop]
- [End of Step 1 If]
14. Exit

Viva Voce

1. **What is DOS?**
2. **Which command is used to create a file in dos?**
3. **What all option do you have in control panel?**
4. **How to move a folder?**
5. **What is linux?**
6. **List any 5 commands of linux?**
7. **What is ms word used for?**
8. **What is the extension of ms word file?**
9. **How to add header and footer in ms word document?**
10. **What is ms excel used for?**
11. **What is the extension of ms excel file?**
12. **What is ms powerpoint used for?**
13. **What is the extension of ms powerpoint file?**
14. **How can we change the layout of a slide?**
15. **What is ms access used for?**
16. **What is the extension of ms access file?**
17. **What is a database?**
18. **What do you mean by attributes and relation?**
19. **What is c++?**
20. **What are different arithmetic operators?**
21. **What is array?**
22. **What are different types of array?**
23. **What is function?**
24. **What is difference between function prototype and function declaration?**
25. **What is constructor and what is its function?**
26. **How is destructor different from constructor?**
27. **What is class and object?**
28. **How do you declare a class?**
29. **What is difference between function and operator overloading?**
30. **What is inheritance ?what is syntax for declaring derived class**